

DIYA SHETTY

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EDUCATION

North Carolina State University – Raleigh, NC

May 2026

Master of Computer Science (MCS) | GPA: 4.0/4.0

Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Foundations of Data Science, Computer Networks

BMS College of Engineering – Bengaluru, India

May 2024

Bachelor of Engineering (B.E.) – Information Science and Engineering

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, JavaScript, TypeScript, C++, HTML, CSS, R
- **Data & AI:** Pandas, NumPy, Apache Spark, PySpark, TensorFlow, Keras, LangChain, RAG, Agentic AI, Streamlit
- **Cloud & Databases:** AWS (Glue, S3, Redshift), Azure (Digital Twins), PostgreSQL, MySQL, MongoDB, ChromaDB, Docker, Linux, Git
- **Concepts & Tools:** ETL/ELT Pipelines, Dimensional Modeling (Star Schema), Data Modeling, REST APIs, OOP, Agile, CI/CD

WORK EXPERIENCE

Data Analyst Intern · Genpact – New York, USA

Jun 2025 – Aug 2025

- Designed and deployed an AI-powered Market Scope Analysis platform using **Agentic AI**, RAG, and LLM-based orchestration; built autonomous agents for web scraping, document retrieval, and vector search, cutting manual research time by ~60%.
- Architected a paired-agent system in **CrewAI** with retrieval and analysis agents per research area running in parallel, with iterative prompt refinement to eliminate hallucination, enforce source-grounded outputs with inline citations, and standardize formatting for consistent report generation.
- Built an AWS Glue ETL pipeline implementing a star schema (fact/dimension tables) to ingest and transform daily records, including cross-format ingestion (CSV, Parquet, JSON) and schema normalization to enable automated downstream analytics.

Summer Intern · EY – Chennai, India

Mar 2024 – Apr 2024

- Designed an SAP-ERP business blueprint for a simulated retail business across 6 core modules (procurement, planning, manufacturing, sales & distribution); approved by manager for demonstrating end-to-end process understanding.
- Built a multi-module ERP demo application in HTML, CSS, JavaScript, and Node.js, translating the blueprint into functional end-to-end workflows.

Technical Operations Intern · Unfold Consulting – Bangalore, India

Aug 2023 – Nov 2023

- Executed 250+ UAT test cases across dev and production environments, enabling early go-live on select features.
- Identified a critical defect in the production environment, along with 10+ other database and data-upload issues, preventing potential post-launch failures; retested fixes alongside developers to validate resolution before deployment.

PROJECTS

NutriPlan – Full-Stack AI Meal Planning App [Link to code](#)

- Engineered a multi-agent LLM pipeline (LangChain + Gemini) with Pydantic-constrained structured output, guaranteeing schema-valid recipe generation across dietary, budget, and macronutrient constraints.
- Designed a deterministic validation and retry pipeline enforcing calorie, allergy, and dietary-restriction compliance, feeding structured failures back into the LLM context to drive self-correction
- Built a secondary intent-classification agent resolving ambiguous requests against live meal-plan state, driving targeted ingredient substitution and meal-swap regeneration.

Document Q&A – Full-Stack RAG Application [Link to Project](#)

[Link to Demo](#)

- Built a production-deployed full-stack RAG pipeline using LangChain agents and ChromaDB vector search, implementing semantic chunking, per-document retrieval scoping, and prompt-level grounding to restrict LLM responses strictly to retrieved context and suppress hallucination.
- Delivered FastAPI StreamingResponse endpoints and a real-time React/TypeScript chat UI backed by a normalized PostgreSQL schema (via SQLAlchemy ORM), enabling per-document conversation history and full Q&A interaction logging for auditability, containerized with Docker Compose and deployed on Railway and Vercel.

IoT – Azure Cloud Digital Twin [Link to Project](#)

 Best Innovation, Engenius 2024

- Developed a real-time cattle health monitoring system using IoT sensors and regression-based anomaly detection, architecting the data pipeline with Azure Digital Twins, Azure Functions, and Azure Data Factory, delivered via 3 REST endpoints to a React Native app for veterinarians.

Natural Language Processing – IEEE Published [Link to Project](#)

[Link to Paper](#)

- Fine-tuned Bio+Clinical BERT across three clinical NLP tasks (NER, question classification, disease prediction) on datasets up to 4.5M tokens; applied LIME/SHAP to expose hidden failures masked by high aggregate accuracy (99.5%)